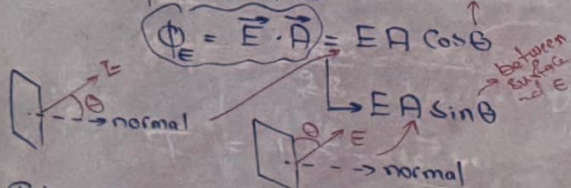


21, 10, 1000

electric flux

(a) open surface



(b) close surface

$$\Phi_E = \frac{Q_{in}}{\epsilon_0} \Rightarrow \Phi_E = \frac{Q_{in}}{\epsilon_0 \epsilon_r}$$

Permittivity constant

for water $\epsilon_r = 80$

$$K = \frac{1}{4\pi\epsilon_0\epsilon_r} = \frac{9 \times 10^9}{80} \Rightarrow K = 0.11 \times 10^9$$

for medium

Magnetic flux

$$\Phi_B = 0$$

why?

because the magnetic field lines are continuous $\rightarrow$ out lines = in lines



out = 2 lines

in = 2 lines

$$\Phi_B = 0$$

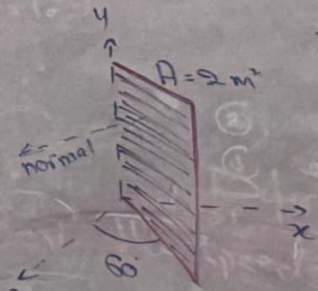
10000 kHz

1 GHz

$q_1 = 2 \mu C$ and $10 \mu C$ separated by 50 cm in water

$$F = k \frac{q_1 q_2}{r^2} = 0.11 \times 10^9 \times \frac{2 \times 10^{-6} \times 10 \times 10^{-6}}{0.5^2}$$

$$F = 0.0088 N = 8.8 mN$$



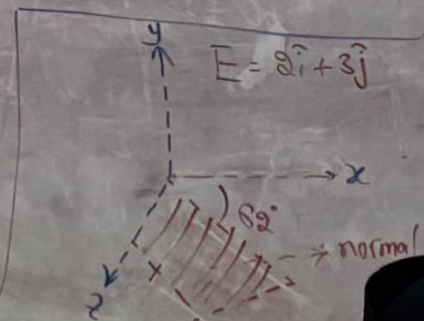
$$\vec{E} = -\hat{i} - 4\hat{j} + 2\hat{k}$$

$$\Phi_E = \vec{E} \cdot \vec{A}$$

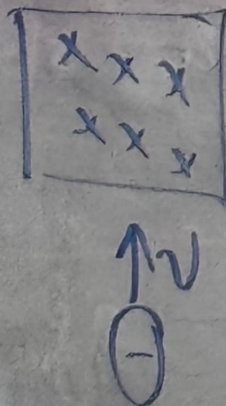
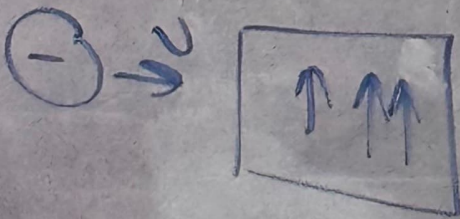
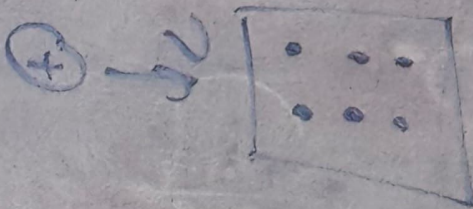
$$\Phi_E = (-\hat{i} - 4\hat{j} + 2\hat{k}) \cdot (-\hat{i} 2 \cos 60^\circ + \hat{k} 2 \sin 60^\circ)$$

$$= 2 \cos 60^\circ + 4 \sin 60^\circ$$

$$= 1 + 2\sqrt{3} = 4.46 N \cdot m^2/C$$

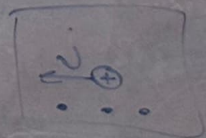
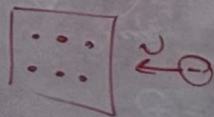
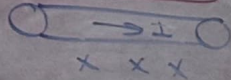
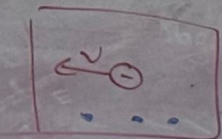


$$T = \frac{2\pi m}{qB}$$



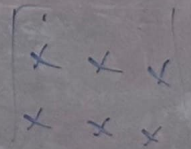
بسم الله الرحمن الرحيم

(c)



$$T = \frac{2\pi m}{qB}$$

$\ominus \rightarrow$



GC enters $B = 2T$ inward, $F_B = 36N$ down

$$v = \frac{F}{qB} = \frac{36}{6 \times 2} = 3 \text{ m/s}$$

- GC moves with $v = 3 \text{ m/s}$ down

$B, F_B = 36N$ inward

$$B = \frac{F}{qv} = \frac{36}{6 \times 3} = 2T$$